

Phase 1: Assisted Development

Module 1: Foundations of AI Collaboration

Building the mindset and skills to effectively collaborate with AI assistants

Agenda

- Introductions
- On the topic of AI — and the question behind the question
- Learning objectives
- Where we're going (the full journey)
- How LLMs work (brief)
- The relationship with your assistant
- Managing the human dynamic
- Model capabilities & limitations
- Balancing control & delegation
- Demonstration & our Phase 1 project

Introductions

- Name
- Experience with AI coding tools
- What you're most curious about



On the Topic of AI

- In general
- In software development
- So then what?

Our Perspective on AI Technology

Realistic:

- Excited about the benefits and precision potential
- Honest about energy costs, computing requirements, and power distribution
- It's a powerful, fun, and technically rich set of innovations
- Change, especially when it comes fast, can be destabilising for people

Our Interest:

- Model selection over defaulting to the most capable/expensive
- Practical domain-specific applications over AGI/ASI speculation
- Smaller, faster, and specialised models

Our Perspective on AI in Software Development

What motivates our belief in this foundational shift:

- The output from AI assistants is **too good to dismiss**
- The sense of productivity gains during implementation is **profound**
- We know we can **increase precision** by applying higher quality context
- The quality of the models is getting **noticeably better, very quickly**

The Question Behind the Question

AI capability isn't in question.

The interesting question is whether we're asking the *right* questions:

- **Can we** automate this? → Yes, often.
- **Should we** automate this? → That requires judgment.

Aristotle called this **phronesis** — practical wisdom. Knowing the right action in a given situation, not just what's technically possible.

Tools That Enhance vs Replace

David Krakauer (Santa Fe Institute) draws a useful distinction:

Tool type	What happens over time
Enhancing	You become more capable through use
Replacing	You outsource the function — and lose the underlying capacity

A map builds your spatial reasoning. GPS replaces it.

The question for AI tools is the same.

This course is designed to put AI tools in the *enhancing* column — for people who use them with intention.

What to Focus On

AI gives you speed. The question is what you do with it.

Flow

- Identify friction and mitigate it — including the friction of losing skills you value
- Flow == intent $\propto$ result

Learning

- Prioritise your learning journey
- Continuous improvement and increased self-awareness

Quality

- What you invest the time and space your assistant gives back to you
- The invitation is not to go faster — it's to go better

Learning Objectives

- **Goal:** Collaborate with an AI assistant to build working code and docs
- **Takeaway:** A working service, built through the collaboration cycle: spec → prompt → plan → implement
- **Hope:** You will have lots of feels 🥹 while remaining curious

Along the way you will:

1. **Experiment** with tools, models, and prompts to develop your own intuition
2. **Recognise** what AI is genuinely good at — and where you need to stay in control
3. **Start building** the collaboration habits the rest of the course builds on

Where We're Going

Module 1 is the first step of a nine-module journey.

Phase	Modules	Focus
Assisted	1–3	Foundations: context, workflow, collaboration
Pairing	4–6	Skills, tools, your pair agent
Agentic	7–9	Agents that work alongside you

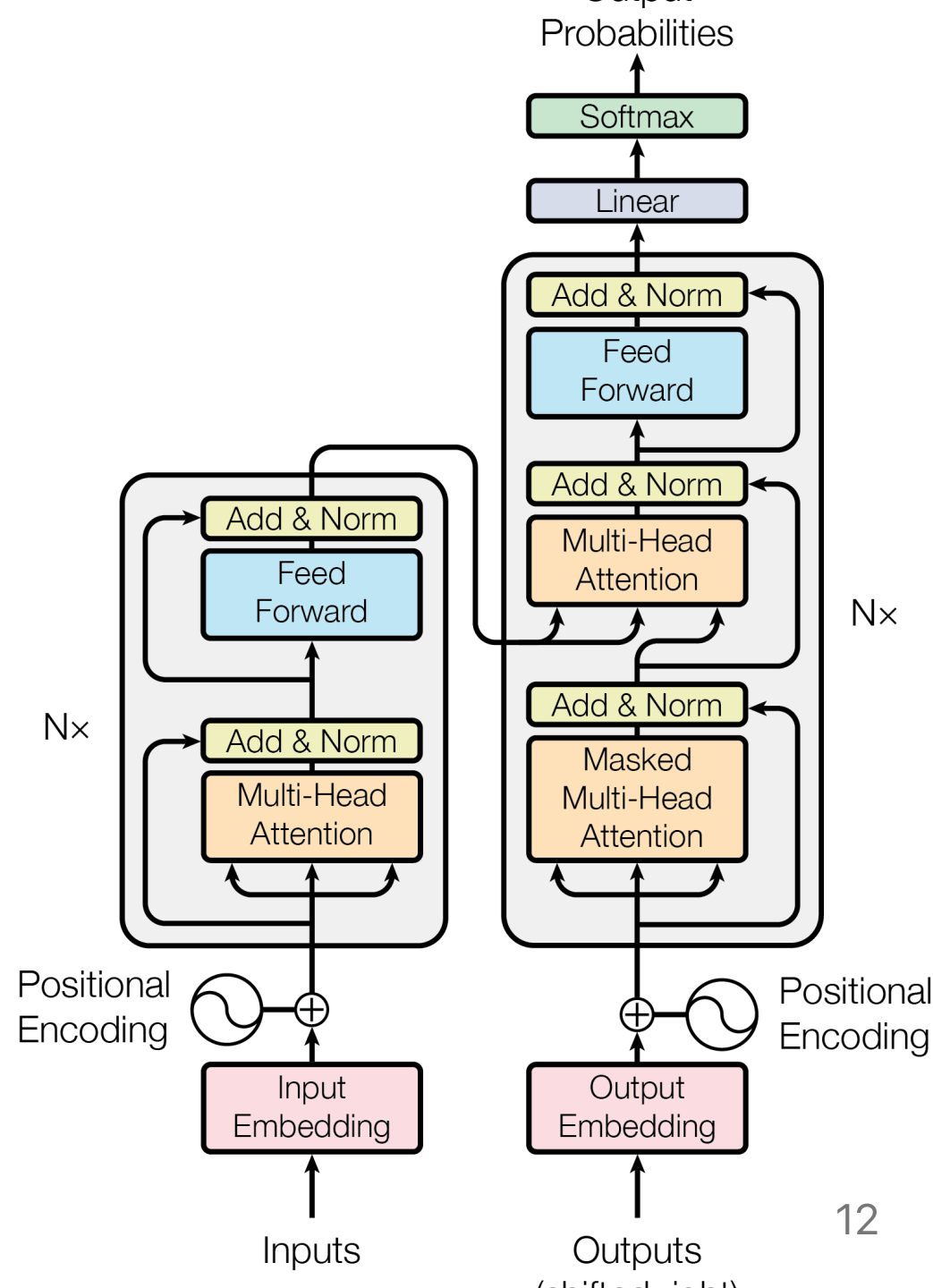
By Module 6 you'll have a **pair agent** — an AI partner that knows your project, follows your workflow, and uses your tools.

Everything in Module 1 is runway for that.

⚙️ How LLMs Work (a discussion)

These terms come up constantly. Let's make sure we have a shared language.

- **Tokens** — what models actually process
- **Training / inference** — how models are built vs. how they're used
- **Context window** — the working memory limit that shapes everything
- **Attention** — the architecture shift that made modern LLMs possible
- **Coding tools** — how these concepts show up in the tools you'll use



AI Roles & Personas

AI assistants act differently depending on the **role** you give them

Role	Purpose
Coder	Writes code from context
Planner	Structures tasks
Architect	Designs systems
Skeptic	Finds flaws and holes in your thinking
Mentor	Explains things you don't understand

The Skeptic and Mentor are the most underused — and often the most valuable.

The *Relationship* with your Assistant

- It *feels* like you're working with a human
 - Familiar, easeful, flexible — using *our* language
- You're just filtering for a *close enough* sequence of tokens
 - Who are you getting upset with?
- Prioritise **yourself** in these interactions
 - Remain kind and curious

Managing the Human Dynamic

You will feel both sides of this

The hard stuff:

- Frustration with repeating imprecise results
- Overwhelm from too many possibilities
- Loss of control and agency

The good stuff:

- Wonder at what just got generated in seconds
- The satisfaction of a plan that actually holds together
- Flow — when intent and result start to align

Strategies: plan first, start small, stay curious, use Git liberally, and keep moving toward flow

Capabilities & Limitations

The point of Modules 1–3 is to help you build this intuition through experience

Current AI is good at

- Code generation and boilerplate (especially when given an example)
- Refactoring, pattern matching, consistency checks, and translation
- Documentation, explanation, and dictation

Common failure modes

- Context window limitations
- Hallucinations (e.g. non-existent APIs)
- Limited business domain understanding

Balancing Control & Delegation

Maintain Control Over

- Architectural decisions
- Business logic validation
- Quality standards
- Final implementation choices

Delegate to AI

- Code scaffolding and boilerplate
- Documentation generation
- Test case suggestions

Key: You remain the decision-maker and are ultimately responsible

A Demonstration

- Setting up your AI assistant and API keys
- Tool and model choice (and why we don't lock you into one)
- See how context matters with different prompts

Project Progression: Module 1

The Config API Service

- Collaborating on the initial plan
- Basic CRUD API for configuration values
- Simple data persistence and validation
- API documentation and testing fundamentals
- Initial experiences of flow, friction, and quality adherence

Let's Go!

Remember:

- You are *still* ultimately responsible/accountable
- Engage the many personas of your assistant
- Focus on learning the collaboration patterns
- Non-attachment to initially generated output
- Be aware of *lazy* tendencies

Answer: How can my assistant help me to optimise **flow, learning, and quality**?

First step: Tool and model selection, and project setup